Course Project Proposal: Research Project

Project Name: Comparative Study of Explainable AI Models for Pneumonia Detection from Chest X-rays

Research Motivation:

I want to conduct a research study on how different deep learning models perform when detecting pneumonia from chest X-ray images and how explainability methods can help understand their predictions. Instead of focusing only on whether a model predicts Pneumonia or Normal, the study will examine prediction performance, confidence scores, and Grad-CAM visual explanations. I am particularly interested in studying cases where two models produce different predictions or focus on different areas of an X-ray. This research will help me understand the strengths, differences, and limitations of the selected models.

Research Methodology:

I will conduct the study using Python, a public chest X-ray dataset, and two pretrained deep learning models. The same dataset and evaluation process will be used for both models so their results can be compared consistently. I will evaluate the models using accuracy, precision, recall, F1-score, and ROC-AUC. Grad-CAM will be used to generate visual explanations for selected predictions, and I will examine cases where the models disagree or produce different explanation patterns. A small Streamlit interface may be developed to help visualize predictions and Grad-CAM results during the research, but the primary focus of the project will be the experimental comparison and analysis.

Expected Research Outcomes:

I expect the study to provide a comparison of how the selected deep learning models perform on pneumonia detection and how their explanations differ. I will analyze situations where the models agree and disagree and examine whether their Grad-CAM explanations highlight similar or different regions of the chest X-ray. The results will help identify differences in model performance and explainability rather than simply producing a working prediction application. The final outcome will be a set of experimental results, comparisons, visualizations, and observations based on the selected dataset and models.

Research Resources:

I will need a computer with sufficient resources to run the pretrained models, a public chest X-ray dataset, and Python libraries such as PyTorch or TensorFlow, OpenCV, NumPy, and Matplotlib. I may also use Streamlit to visualize selected experimental results. The main resources required will be the dataset, pretrained models, computing resources, and time for model evaluation, Grad-CAM analysis, and comparison of experimental results.

I used Microsoft Copilot as the LLM while developing this project idea.

Prompt 1

I am interested in conducting a research study using Python to analyze chest X-ray images for signs of pneumonia. I want to compare deep learning models and study not only their predictions but also why they make those predictions. I am interested in using Grad-CAM to visually examine the areas of the X-ray that influence model predictions. How could I design an experimental methodology for this research study?

Copilot Response 1

The research could use a public chest X-ray dataset and two pretrained deep learning models to perform pneumonia classification. The same test images could be evaluated by both models, and their predictions and confidence scores could be recorded for comparison. Grad-CAM could then be applied to selected predictions to visualize the regions that influenced each model. The study could analyze model performance using standard evaluation metrics and examine whether the models focus on similar or different regions when making predictions.

Prompt 2

I also want to investigate how the two models differ in their performance and visual explanations. How could I design the analysis to identify model disagreements and determine whether the models focus on different regions of chest X-ray images when making their predictions?

Copilot Response 2

The study could evaluate both models on the same test dataset and compare their accuracy,

precision, recall, F1-score, and ROC-AUC. Cases where the models produce different predictions could be identified for further analysis. Grad-CAM explanations could be generated for these cases and compared visually to determine whether the models focus on similar or different regions of the X-ray. The research could summarize these findings using tables, performance comparisons, and representative visual examples.